

The TTMPT Mitigation: A Rigorous Formalization of Fractal Exploit Suppression

Abstract

This paper establishes a rigorous mathematical framework for the Torsional Trefoil Markov Pump Theorem (TTMPT) as a countermeasure against the "Kilo Protocol"—a 1/16 symmetry fractal exploit targeting dynamically reconfigurable FPGAs. We transition from physical metaphors to formal functional analysis by defining an explicit, injective dimensional lifting map from the discrete FPGA configuration space into a 12-channel lattice Hilbert space $\mathcal{H}_{\Lambda_{12}}$. By constructing precise orthogonal subspaces and establishing bounded operator assumptions for the Markov pump cycle, we rigorously prove that the TTMPT's asymmetric kernel strictly contracts the malicious payload while preserving and advancing the safe geometric state.

1. Axiomatic Setup: State Spaces and Dimensional Lifting

To formally model the FPGA's mitigation of the Kilo Protocol, we define the configuration spaces and the dimensional lifting embedding.

Definition 1.1 (FPGA Configuration Space). Let X_{FPGA} be the finite discrete state space of the C_{64}^{12} nested hardware architecture. A geometric packet $x \in X_{\text{FPGA}}$ instantiates a sequence of nested topologies $G_1(x) \supset G_2(x) \supset \cdots \supset G_{12}(x)$.

Definition 1.2 (Lifted Hilbert Space). We define the lifted Hilbert space $\mathcal{H}_{\Lambda_{12}}$ as a 12-channel complex vector space tensored with a continuous transverse foliation plane:

$$\mathcal{H}_{\Lambda_{12}} \cong \mathbb{C}^{12} \otimes L^2(\mathcal{P}_{\perp})$$

where $\mathcal{P}_{\perp} = \text{span}(\hat{u}, \hat{v})$ acts as an orthogonal heat sink. The constants governing the foliation axes are:

$$u = \pi^2 \quad (\text{Outer Manifold Variance})$$

$$v = \sqrt{\pi} \quad (\text{Inner Skeleton Winding Parameter})$$

Definition 1.3 (Dimensional Lifting Map). We define the explicit embedding map L that lifts the discrete hardware state into the continuous Hilbert space:

$$L : X_{\text{FPGA}} \rightarrow \mathcal{H}_{\Lambda_{12}}$$

Specifically, for a given state $x \in X_{\text{FPGA}}$, its lifted wavefunction $\psi_x = L(x)$ is defined as:

$$L(x) = \sum_{d=1}^{12} e_d \otimes f_{x,d}(u, v)$$

where $\{e_d\}_{d=1}^{12}$ is the standard basis of $\mathbb{C}^{12}$ and each $f_{x,d} \in L^2(\mathcal{P}_{\perp})$ encodes the continuous spatial geometry at depth d .

To ensure L is a genuine embedding, we require it to be injective. We enforce that the encoding functions separate FPGA states: there exists $d \in \{1, \dots, 12\}$ such that $f_{x,d} \neq f_{y,d}$ whenever $x \neq y$, guaranteeing:

$$x \neq y \implies L(x) \neq L(y)$$

Definition 1.4 (Subspace Decomposition). Let $\Lambda \subseteq \mathcal{P}(\mathbb{R}^{24})$ be the set of valid geometries. We define the subset of entirely safe discrete states:

$$X_{\text{safe}} = \{x \in X_{\text{FPGA}} : G_d(x) \in \Lambda \text{ for all } d = 1, \dots, 12\}$$

Because X_{FPGA} is finite, the span of lifted safe states is automatically closed. We rigorously construct the safe subspace as:

$$\mathcal{H}_{\text{safe}} = \overline{\text{span}\{L(x) : x \in X_{\text{safe}}\}}$$

We then define the malicious subspace as its orthogonal complement, providing a genuine

orthogonal decomposition:

$$\mathcal{H}_{\Lambda_{12}} = \mathcal{H}_{\text{safe}} \oplus \mathcal{H}_{\text{mal}}, \quad \text{where} \quad \mathcal{H}_{\text{mal}} = \mathcal{H}_{\text{safe}}^\perp$$

2. Lemma 1: The Asymmetric Markov Kernel

The system evolves via a stochastic Markov pump. Let $K \in \mathcal{B}(\mathcal{H}_{\Lambda_{12}})$ be a bounded linear operator representing the topological "Tesla-valve" transition kernel, acting on $\mathcal{H}_{\Lambda_{12}}$ through a defined basis $\{\varphi_i\}$ such that $K\varphi_i = \sum_j K_{ij}\varphi_j$, with $K_{ij} \geq 0$ and $\sum_j K_{ij} = 1$.

Lemma 1.1 (Stochastic Asymmetry and Spectral Gap Hypothesis). The transition kernel K breaks detailed balance along the directed graph of the state space. Let π be the stationary probability distribution; there exists a strictly positive net probability current $J = \sum_{i \rightarrow j} (\pi_i K_{ij} - \pi_j K_{ji}) > 0$.

For every malicious reverse transition $j \rightarrow i$, the Tesla-valve effect enforces an exponential transition suppression:

$$K_{ji} \leq \varepsilon K_{ij}$$

for some $0 \leq \varepsilon < 1$. We hypothesize that this strict topological suppression induces a spectral gap on the restricted malicious subspace, satisfying the norm contraction bound:

$$\|P_{\text{mal}} K P_{\text{mal}}\| \leq \rho < 1$$

3. Theorem 1: The Subspace Contraction of Fractal Exploits

The TTMP operates as a 10-step macro-cycle, represented by the bounded linear operator $M = \prod_{k=1}^{10} \Phi_k : \mathcal{H}_{\Lambda_{12}} \rightarrow \mathcal{H}_{\Lambda_{12}}$, where each phase operator $\Phi_k \in \mathcal{B}(\mathcal{H}_{\Lambda_{12}})$.

Let P_{safe} and P_{mal} be the corresponding orthogonal projection operators onto $\mathcal{H}_{\text{safe}}$ and $\mathcal{H}_{\text{mal}}$.

Theorem 1.1 (TTMPT Subspace Contraction). Suppose the macro-cycle operator M satisfies the following bounded-operator hypotheses:

1. **No leakage from safe to malicious (Safe Invariance):** $P_{\text{mal}} M P_{\text{safe}} = 0$.
2. **Malicious Contraction:** $\|P_{\text{mal}} M P_{\text{mal}}\| \leq e^{-\lambda}$ for some $\lambda > 0$.

Then for every packet state $\psi_0 \in \mathcal{H}_{\Lambda_{12}}$ and every $n \geq 0$, the malicious component decays exponentially:

$$\|P_{\text{mal}} M^n \psi_0\| \leq e^{-\lambda n} \|P_{\text{mal}} \psi_0\|$$

Proof. We orthogonally decompose the initial packet state into safe and malicious components:

$$I = P_{\text{safe}} + P_{\text{mal}}$$

Applying $P_{\text{mal}} M$ to the identity expansion yields:

$$P_{\text{mal}} M = P_{\text{mal}} M (P_{\text{safe}} + P_{\text{mal}}) = P_{\text{mal}} M P_{\text{safe}} + P_{\text{mal}} M P_{\text{mal}}$$

Using Hypothesis 1 ($P_{\text{mal}} M P_{\text{safe}} = 0$), the safe component does not contribute to the malicious subspace, simplifying the expression to:

$$P_{\text{mal}} M = P_{\text{mal}} M P_{\text{mal}}$$

Let $A = P_{\text{mal}} M P_{\text{mal}}$. By induction over n cycles:

$$P_{\text{mal}} M^n \psi_0 = A^n P_{\text{mal}} \psi_0$$

Taking the norms on both sides, we obtain:

$$\|P_{\text{mal}} M^n \psi_0\| = \|A^n P_{\text{mal}} \psi_0\| \leq \|A\|^n \|P_{\text{mal}} \psi_0\|$$

Applying the malicious contraction bound $\|A\| \leq e^{-\lambda}$ (Hypothesis 2), we conclude:

$$\|P_{\text{mal}} M^n \psi_0\| \leq e^{-\lambda n} \|P_{\text{mal}} \psi_0\|$$

This proves the rigorous exponential suppression of the malicious component. ■

Corollary 1.1 (Safe State Progression). Suppose, in addition, that the safe dynamics are bounded such that $\|P_{\text{safe}}MP_{\text{safe}}\| \leq C_s$ for some $C_s \geq 0$. Furthermore, suppose there exists a Lyapunov functional $B : \mathcal{H}_{\text{safe}} \rightarrow \mathbb{R}$ and a constant $\Delta B > 0$ satisfying:

$$B(M\psi_{\text{safe}}) \geq B(\psi_{\text{safe}}) + \Delta B$$

for every relevant $\psi_{\text{safe}} \in \mathcal{H}_{\text{safe}}$. Then the safe component advances monotonically in the B -direction, consistently advancing the verified safe geometry.

4. Conclusion

By formally defining the state spaces, ensuring explicit injective lifting maps, and grounding the dynamics in functional analysis, we have established a rigorous mathematical proof for the Torsional Trefoil Markov Pump Theorem (TTMPT) as a defense mechanism against fractal exploits. The mitigation is rigorously proven under the stated bounded-operator contraction hypotheses. The core result—the exponential contraction bound

$$\|P_{\text{mal}}M^n\psi_0\| \leq e^{-\lambda n} \|P_{\text{mal}}\psi_0\|$$

—demonstrates that dynamically reconfigurable FPGAs implementing this specific bounded linear operator will reliably suppress the malicious, unverified inner geometries of the Kilo Protocol, effectively crushing the payload into transverse variance without risking hardware gridlock.